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Thermal Transport across Ion-cut Monocrystalline β-Ga_2O_3 Thin Films and Bonded β-Ga_2O_3-SiC Interfaces

Zhe Cheng[1,a)], Fengwen Mu[2,4,a),*], Tiangui You[3,a)], Wenhui Xu[3], Jingjing Shi[1], Michael E. Liao[5], Yekan Wang[5], Kenny Huynh[5], Tadatomo Suga[4], Mark S. Goorsky[5], Xin Ou[3,*], Samuel Graham[1,6,*]

[1] George W. Woodruff School of Mechanical Engineering, Georgia Institute of Technology, Atlanta, Georgia 30332, USA

[2] Kagami Memorial Research Institute for Materials Science and Technology, Waseda University, Shinjuku, Tokyo 169-0051, Japan

[3] State Key Laboratory of Functional Materials for Informatics, Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, 200050, China

[4] Collaborative Research Center, Meisei University, Hino-shi, Tokyo 191-8506, Japan

[5] Materials Science and Engineering, University of California, Los Angeles, Los Angeles, CA, 90095, United States

[6] School of Materials Science and Engineering, Georgia Institute of Technology, Atlanta, Georgia 30332, USA

[a)] These authors contributed equally

[*] Corresponding authors: mufengwen123@gmail.com; ouxin@mail.sim.ac.cn; sgraham@gatech.edu

ABSTRACT

The ultra-wide bandgap, high breakdown electric field, and large-area affordable substrates make β-Ga_2O_3 promising for applications of next-generation power electronics while its thermal conductivity is at least one order of magnitude lower than other wide/ultrawide bandgap semiconductors. To avoid the degradation of device performance and reliability induced by the localized Joule-heating, proper thermal management strategies are essential, especially for high-power high-frequency applications. This work reports a scalable thermal management strategy to heterogeneously integrate wafer-scale monocrystalline β-Ga_2O_3 thin films on high thermal conductivity SiC substrates by ion-cutting technique and room-temperature surface-activated bonding technique. The thermal boundary conductance (TBC) of the β-Ga_2O_3-SiC interfaces and thermal conductivity of the β-Ga_2O_3 thin films were measured by Time-domain Thermoreflectance (TDTR) to evaluate the effects of interlayer thickness and thermal annealing. Materials characterizations were performed to understand the mechanisms of thermal transport in these structures. The results show that the β-Ga_2O_3-SiC TBC values are reasonably high and increase with decreasing interlayer thickness. The β-Ga_2O_3 thermal conductivity increases more than twice after annealing at 800 ^oC due to the removal of implantation-induced strain in the films. A Callaway model is built to understand the measured thermal conductivity. Small spot-to-spot variations of both TBC and Ga_2O_3 thermal conductivity confirm the uniformity and high-quality of the bonding and exfoliation. Our work paves the way for thermal management of power electronics and provides a platform for β-Ga_2O_3 related semiconductor devices with excellent thermal dissipation.

KEYWORDS: Gallium Oxide; thermal management; ion cut; surface activated bonding; TDTR; thermal boundary conductance; SiC

INTRODUCTION

Due to the ultra-wide bandgap (4.8 eV), high breakdown electric field (8 MV/cm), and large-area affordable substrates grown from melt, β-Ga_2O_3 has attracted great attention for potential applications of next-generation power electronics recently.[1-4] The Baliga figure of merit of β-Ga_2O_3 is several orders of magnitude higher than that of Si.[3, 5-6] However, the thermal conductivity of bulk β-Ga_2O_3 is at least one order of magnitude lower than those of other wide bandgap semiconductors and is highly anisotropic due to the low (monoclinic) crystal structure.[7-10] Moreover, the thermal conductivity of β-Ga_2O_3 nanostructures, such as thin films, superlattice, and nanocrystalline films, is further reduced significantly compared with that of the bulk β-Ga_2O_3.[7, 11-12] For high-frequency and high-power applications, the device performance and reliability are limited by the high channel temperature induced by the localized Joule-heating near the gate. Therefore, proper, even aggressive, thermal management strategies are of great importance to avoid device degradation.[4] However, compared with current demonstrations of β-Ga_2O_3 devices, a disproportionately small amount of work is addressing this sensitive issue.[4]

To extract the heat out of power electronic devices, high thermal conductivity substrates or top metal contacts with high thermal boundary conductance (TBC) near the gate are favorable.[10, 13-14] However, direct growth of high-quality β-Ga_2O_3 on high thermal conductivity substrates such as diamond and SiC is difficult due to the differences in crystal symmetry between β-Ga_2O_3 and substrates.[12] Recently, devices based on mechanically exfoliated monocrystalline β-Ga_2O_3 nano-membranes have been fabricated and a record high drain current has been achieved via van der Waals bonding with single crystalline diamond.[15-19] These works demonstrated the great potential of β-Ga_2O_3 devices. However, these devices have limited capability to scale up to wafer-scale

fabrications. Moreover, the van der Waals bonding leads to a low TBC between β-Ga_2O_3 and the substrates (17 MW/m^2-K), which hinders the heat spreading/extraction capability of the high thermal conductivity substrates.[11]

In this work, we report a scalable wafer-scale strategy to heterogeneously integrate nanoscale monocrystalline β-Ga_2O_3 thin films on high thermal conductivity SiC substrates via ion-cutting and surface-activated bonding techniques. Time-domain thermoreflectance (TDTR) is used to measure the thermal conductivity of the β-Ga_2O_3 thin films and the TBC of the β-Ga_2O_3-SiC interfaces. Detailed materials characterizations such as a transmission electron microscopy (TEM), high resolution X-ray diffraction (HRXRD), and electron energy loss spectroscopy (EELS) are performed to study the structure-property relation of the β-Ga_2O_3 thin films and the interfaces. Additionally, we build some thermal models to understand the measured thermal conductivity and TBC.

RESULTS AND DISCUSSION

Four samples (10 mm x 8 mm) were prepared in this work, as shown in the schematic diagram of Figure 1. 2-inch bulk ($\bar{2}$01) β-Ga_2O_3 wafers were implanted with hydrogen ions. A layer of Al_2O_3 is deposited on the β-Ga_2O_3 wafers by atomic layer deposition (ALD) before bonding to a 4-inch 4H-SiC wafer at room temperature via the surface activated bonding technique. The Al_2O_3 layers might be able to block the electrical breakdown due to the interface defects in devices while they induce additional thermal resistance. After that, a thin layer of monocrystalline β-Ga_2O_3 was exfoliated from the bulk β-Ga_2O_3 wafer by heating up to 450 ^oC. The bulk β-Ga_2O_3 wafers can be used for multiple exfoliations, which save materials and lower the fabrication cost. More details

about the sample preparation can be found in the Methods section. We name the four samples as Samp1, Samp2, Samp3, and Samp4. Samp1 and Samp2 have a 30-nm Al_2O_3 interlayer while Samp3 and Samp4 have a 10-nm Al_2O_3 interlayer. Samp1 and Samp3 are as-bonded while Samp2 and Samp4 were annealed at 800 °C in N_2.

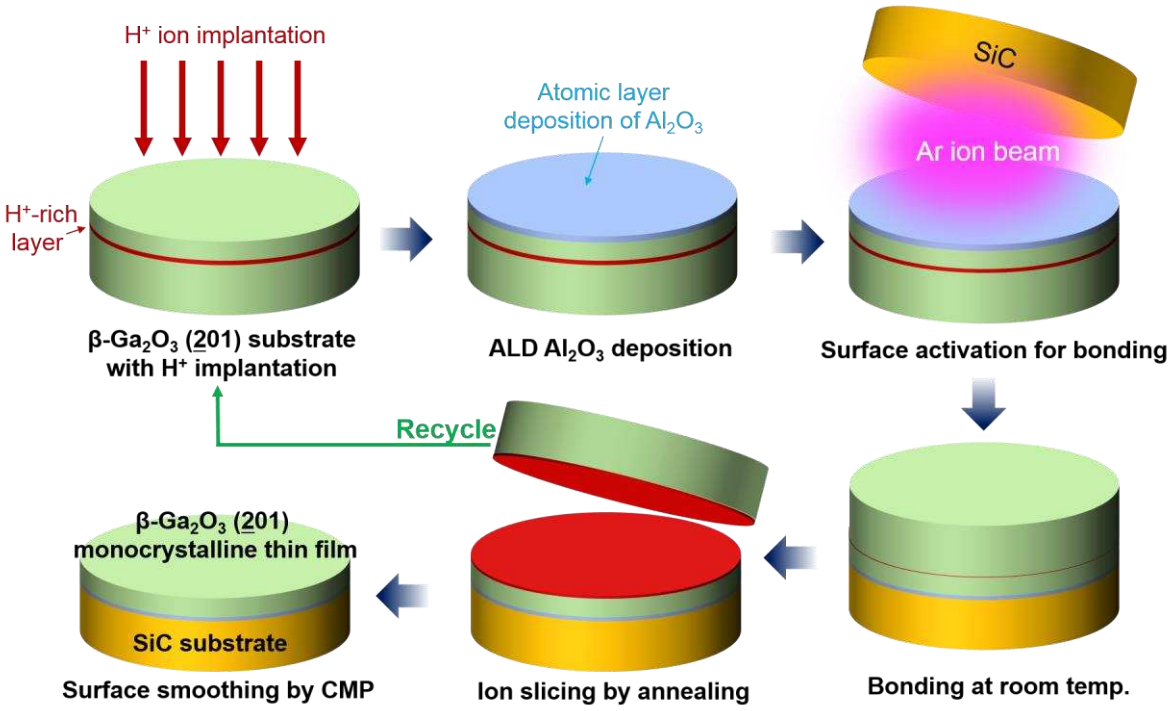


Figure 1. Schematic diagram of sample preparation. A layer of Al_2O_3 is deposited on the β-Ga_2O_3 wafer which is implanted by H ions. After that, surface activated bonding is used to bond the Al_2O_3 coated β-Ga_2O_3 wafer with a 4H-SiC wafer at room temperature. A thin layer of monocrystalline β-Ga_2O_3 is transferred on the SiC wafer after heating up to 450 °C. The surface can be smoothed by chemical mechanical polishing (CMP) and the rest of the β-Ga_2O_3 wafer can be used repeatedly.

Figure 2 (a) shows a 2-inch (2̲01) β-Ga_2O_3 wafer bonded with a 4-inch 4H-SiC wafer. We can see the excellent quality of the wafer bonding. No voids are observed in the bonded area. Our work brings the bulk-quality monocrystalline (2̲01) β-Ga_2O_3 directly bonded to high thermal

conductivity SiC. Figure 2(b) shows the image of the monocrystalline (2̲01) β-Ga_2O_3 transferred onto the 4H-SiC substrate. Only a monocrystalline nanoscale thin film (~140 nm thick) remains on the SiC wafer, which allows for later device fabrication or homo-epitaxial growth of β-Ga_2O_3 structures on these β-Ga_2O_3-on-SiC wafers. We did not see any voids on these exfoliated β-Ga_2O_3 thin films, which confirms the high-quality bonding and exfoliation. Furthermore, the bulk (2̲01) β-Ga_2O_3 wafer can be repeatedly used for multiple exfoliations.

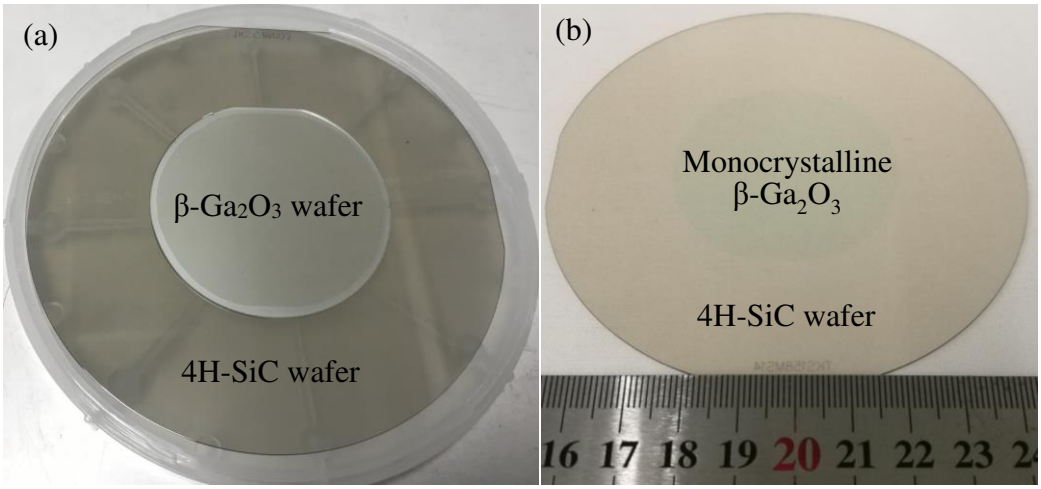


Figure 2. (a) A 2-inch β-Ga_2O_3 wafer is bonded on a 4-inch 4H-SiC wafer. (b) The monocrystalline β-Ga_2O_3 thin film is transferred onto the 4H-SiC wafer.

To evaluate the heat dissipation capability of these β-Ga_2O_3-on-SiC wafers, we measured the thermal boundary conductance (TBC) of the β-Ga_2O_3-SiC interfaces and the thermal conductivity of the β-Ga_2O_3 thin films by TDTR. TDTR is an optical pump-probe technique for thermal characterization of both bulk and nanostructured materials.[9, 20-22] More details about the TDTR measurements can be found in the Methods section. Here, we will discuss the β-Ga_2O_3-SiC TBC first. As shown in Figure 3(a), the samples have three layers: Al transducer, β-Ga_2O_3 layer and SiC substrate. Here, the β-Ga_2O_3-Al_2O_3-SiC TBC is an effective TBC which considers the thermal

resistance of the β-Ga_2O_3-Al_2O_3 interface, the Al_2O_3 layer, and the Al_2O_3-SiC interface: $1/TBC(\beta\text{-}Ga_2O_3\text{-}Al_2O_3\text{-}SiC)=1/TBC(\beta\text{-}Ga_2O_3\text{-}Al_2O_3)+R(Al_2O_3)+1/TBC(Al_2O_3\text{-}SiC)$. Here, $R(Al_2O_3)$ is the thermal resistance of the Al_2O_3 layer. The TBC of the four samples were measured to evaluate the effects of Al_2O_3 thickness and annealing. Figure 3(b) shows the β-Ga_2O_3-SiC TBC of Samp1 (30 nm Al_2O_3 interlayer, as-bonded) and Samp2 (30 nm Al_2O_3 interlayer, annealed at 800 ^oC in N_2). 14 arbitrarily selected spots were measured on each sample to check the uniformity of the bonding. The TBC variations are less than 6%, which confirms the uniformity of the bonded interfaces. Figure 3(c) shows the measured β-Ga_2O_3-SiC TBC of Samp3 (10 nm Al_2O_3 interlayer, as-bonded) and Samp4 (10 nm Al_2O_3 interlayer, annealed at 800 ^oC in N_2). Uniform TBC is also observed for four different randomly selected spots on the samples. The β-Ga_2O_3-SiC TBC of Samp3 and Samp4 are larger than those of Samp1 and Samp2 with thicker Al_2O_3 interlayers. Al_2O_3 interlayers induce additional thermal resistances at the β-Ga_2O_3-SiC interfaces so thinner interlayers lead to smaller thermal resistances and larger effective TBC. The TBC of the as-bonded, unannealed samples are slightly larger than those of the annealed interfaces. It is notable that the breakdown electric field of ALD Al_2O_3 decreases with increasing thickness.[23] The breakdown electric field of ALD Al_2O_3 may be smaller than that of β-Ga_2O_3 with large thickness.[23-24] Therefore, thinner ALD Al_2O_3 interlayers are better for devices in terms of both thermal and electric performances. Directly bonded β-Ga_2O_3-SiC interfaces without Al_2O_3 interlayers will be reported in our future work.

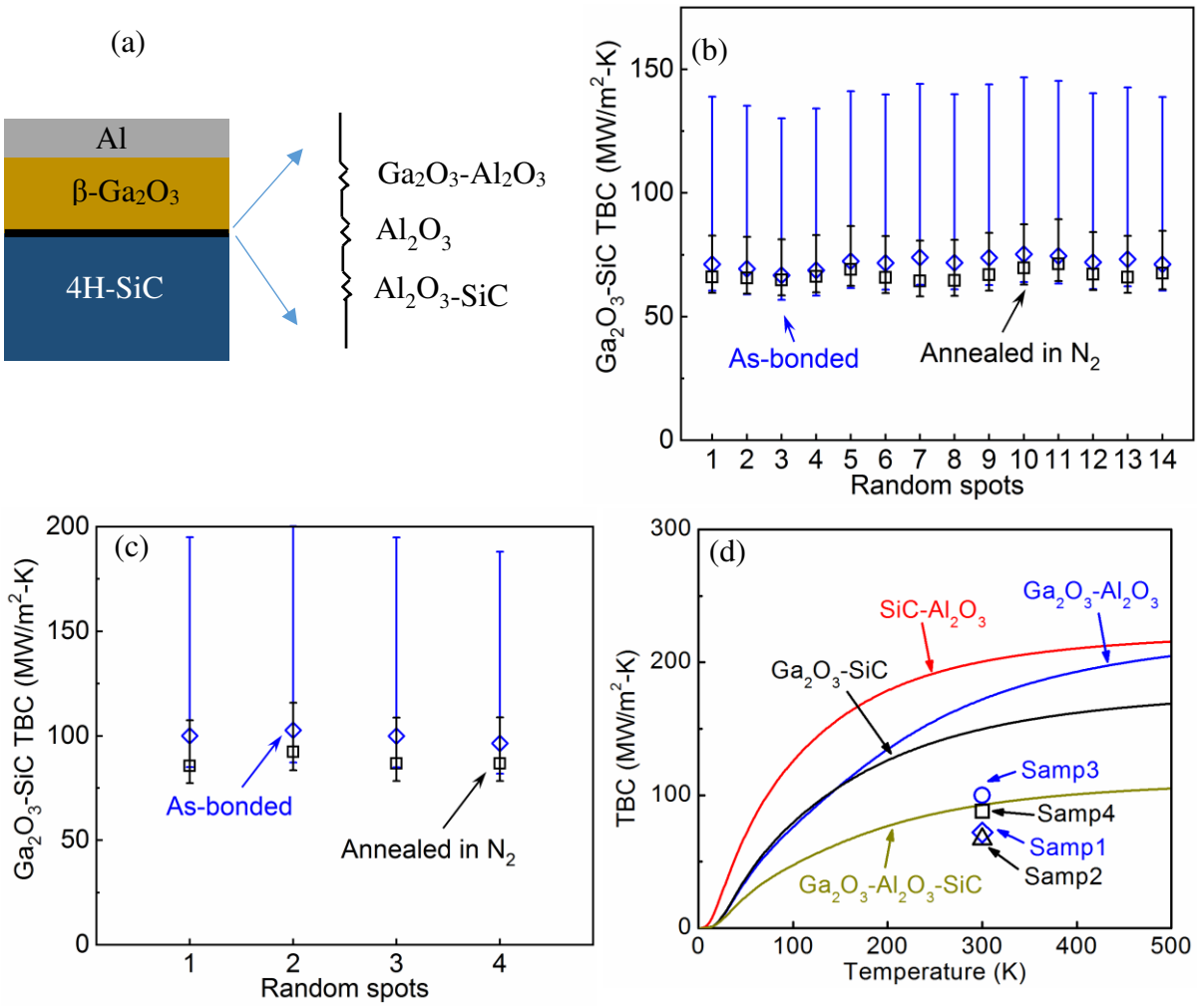


Figure 3. (a) The schematic diagram of the sample structure and thermal resistances. (b) The measured β-Ga_2O_3-SiC TBC of Samp1 and Samp2 which have a 30-nm Al_2O_3 interlayer. 14 different spots were measured to show the uniformity of the samples. (c) The measured β-Ga_2O_3-SiC TBC of Samp3 and Samp4 which have a 10-nm Al_2O_3 interlayer. (d) Comparison of measured TBC and calculated TBC.

Figure 3(d) shows the comparison of the measured TBC of the four samples and the calculated temperature dependent TBC of the β-Ga_2O_3-SiC interfaces, the β-Ga_2O_3-Al_2O_3 interfaces, and the Al_2O_3-SiC interfaces. Density-functional-theory (DFT) is used to calculate the phonon dispersion

relation of the β-Ga_2O_3, Al_2O_3, and 4H-SiC. Diffuse mismatch model (DMM) is used to calculate the phonon transmission and the TBC can be evaluated by the Landauer Formula.[11, 25] The detailed derivation and formula used for this calculation can be found in the Methods section. The β-Ga_2O_3-SiC and β-Ga_2O_3-Al_2O_3 TBC are not very large due to the relatively small phonon group velocity (6620 m/s) of β-Ga_2O_3 in the direction perpendicular to the ($\underline{2}$01) plane.[11, 26] As temperature increases, an increasing number of phonons are excited. The increased TBC with increased temperature is due to the increased number of phonons involved interfacial thermal transport. We see good agreement between the measured TBC and calculated TBC at room temperature. This demonstrates the capability of surface activated bonding to achieve thermally conductive interfaces as predicted by theoretical models.

To understand the measured TBC, TEM is used to characterize the interfacial structure. Figure 4(a-b) shows the interfacial structures of Samp1 and Samp2. The β-Ga_2O_3 is very well-bonded with the SiC wafer without any voids. The β-Ga_2O_3-Al_2O_3 interfaces are very smooth and both sides of these interfaces are crystalline materials, while there are amorphous layers at the Al_2O_3-SiC interfaces. These amorphous layers result from the surface activation by Ar ions during the surface activated bonding process. By comparing Samp1 and Samp2, the annealing process makes the amorphous layers slightly thinner (from ~3.5 nm to ~2 nm), which facilitates thermal transport across interfaces. Also, annealing can possibly improve the crystalline quality of the Al_2O_3 interlayers, which facilitates thermal transport as well. However, we observed a slightly reduced TBC, especially for Samp4.

To figure out the potential mechanisms of the slightly reduced TBC after annealing (comparison of Samp3 and Samp4), STEM and EELS were performed to study the interfacial structures of Samp3 and Samp4, as shown in Figures 4(c-f). Figures 4 (c-d) show the STEM bright-field (BF) images of the β-Ga_2O_3-SiC interfaces of Samp3 and Samp4, respectively. The interfaces consist of ~9-nm-thick ALD-Al_2O_3 layers and ~2-nm amorphous SiC layers. Similar to the amorphous layers in Samp1 and Samp2, these amorphous layers result from the surface activation by the Ar ion beams. The ions impact the SiC surface and break the bonds near the surface, which turns the crystalline SiC into amorphous SiC. Figures 4 (e-f) show the EELS mapping of Ga element distributions of the interface of Samp3 and Samp4. Ga element or GaO_x diffusion into the Al_2O_3 interlayers to form AlGaO alloys is observed after annealing in N_2 gas at 800 ^oC. Alloy structure scatters phonons extensively and reduce thermal conductivity. We attribute this alloy effect as the possible reason for the slightly reduced TBC measured by TDTR.

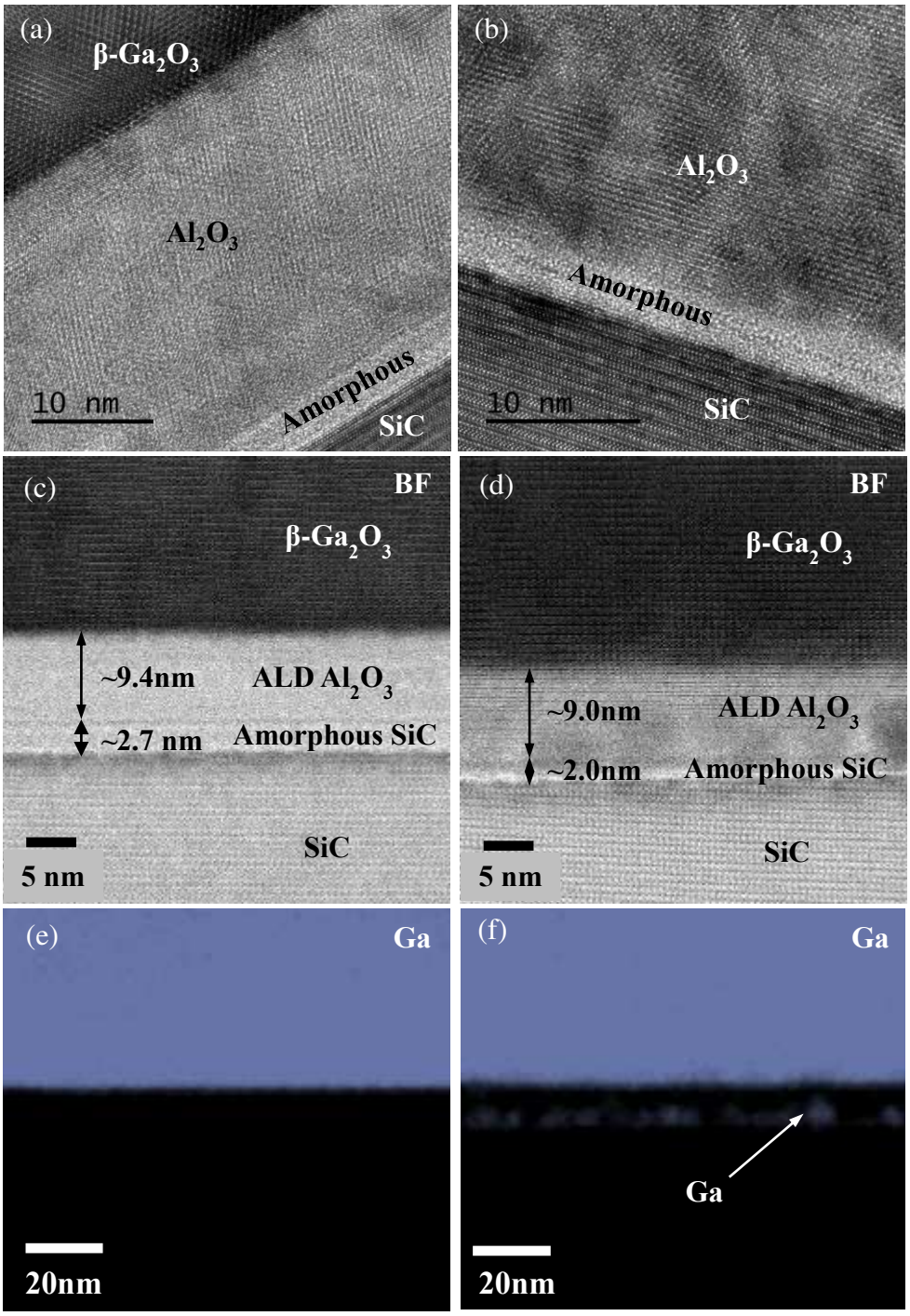


Figure 4. (a) HRTEM images of Samp1 with a 30-nm Al_2O_3 interlayer (as-bonded). (b) HRTEM images of Samp2 with a 30-nm Al_2O_3 interlayer (annealed). (c) STEM images of the interface of Samp3 with a 10-nm Al_2O_3 interlayer (as-bonded). (d) STEM images of the interface of Samp4 with a 10-nm Al_2O_3 interlayer (annealed). (e) Ga element EELS map of the interface of Samp3.

(f) Ga EELS map of the interface of Samp4. Ga diffusion is observed after annealing in N_2 at 800 ^oC.

The thermal conductivity of the monocrystalline nanoscale β-Ga_2O_3 thin films were also measured by TDTR, as shown in Figure 5. Figure 5(a) shows the measured thermal conductivity of β-Ga_2O_3 thin films of Samp1 and Samp2 while Figure 5(b) shows those of Samp3 and Samp4. 14 random spots were measured to check the uniformity of the β-Ga_2O_3 thin films. The small variation of the measured thermal conductivity confirms the uniformity of these thin films. The measured thermal conductivity of the monocrystalline ($\bar{2}$01) β-Ga_2O_3 thin films (2.9 W/m-K) is more than four times lower than the bulk counterpart (13.3 W/m-K).[26] The significant reduction in thermal conductivity is due to the film boundary scattering and the implantation-induced strain in the thin films, which will be discussed in detail later.[27] The thermal conductivity of the β-Ga_2O_3 thin films increases more than twice after annealing in N_2 at 800 ^oC. Annealing reduces the implantation-induced strain in the thin films. As shown in Figure 5(b), a similar increase in the thermal conductivity of the β-Ga_2O_3 thin films was observed Samp4. There are some slight thickness variations of the β-Ga_2O_3 thin films for these four different samples.

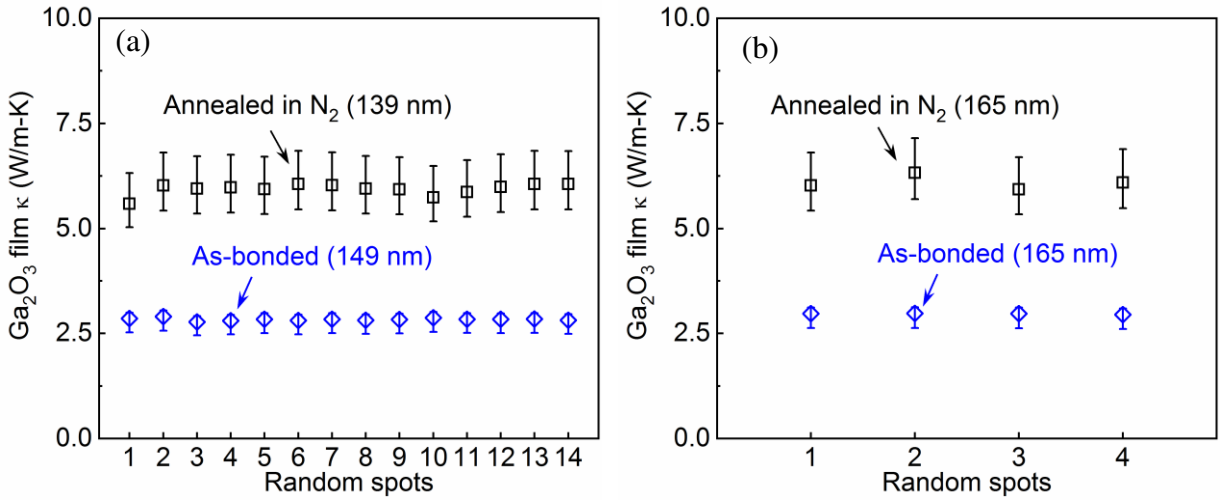


Figure 5. (a) The measured thermal conductivity of monocrystalline β-Ga_2O_3 thin films of Samp1 and Samp2. (b) The measured thermal conductivity of monocrystalline β-Ga_2O_3 thin films of Samp3 and Samp4.

To evaluate the strain variation before and after annealing in N_2, we performed triple-axis XRD measurements on Samp1 and Samp2, as shown in Figure 6(a-b). The hydrogen implantation elastically distorts the lattice through the introduction of point defects (lattice vacancies, knocked-on Ga and O, intercalated H) and can introduce measurable strain.[28-31] Figure 6(a) is the triple-axis diffraction and the peaks are relative to the (0004) 4H-SiC. The peaks near ~ -30,000 arcsec are from the β-Ga_2O_3 layers; this large offset shows the angular difference between the ($\underline{2}$01) β-Ga_2O_3 peak and the (0004) 4H-SiC peak. Samp1 shows an asymmetric peak and is shifted to a lower angle (larger out-of-plane lattice parameter as expected for implanted material) from the expected lattice parameter of β-Ga_2O_3, which indicates that the strain from the hydrogen implantation is not completely removed for the as-bonded samples. The fringe period corresponds to a thickness of ~140 nm, which agrees with the thicknesses measured by TEM and picosecond acoustic technique. Samp2 shifts to a smaller angular separation expected for ($\underline{2}$01) β-Ga_2O_3. Indeed, the position of

the peak after annealing corresponds to an interplanar spacing of 0.4684 nm which matches that known for bulk β-Ga_2O_3 (0.4680 nm). In addition, the peak is much more symmetric with similar intensity fringes on either side of the main peak, as expected for a layer whose implant strain-induced elastic distortions have been relieved. Thus, all of the implant-induced strain is removed from the samples annealed at 800 ^oC in N_2. This is similar to what has been observed in other implanted materials,[27-28] particularly those used for exfoliation.[30-31] The fringe period also corresponds to ~140 nm in agreement with the TEM analysis.

Figure 6(b) shows the triple-axis rocking curves (ω scans) of Samp1 (as-bonded) and Samp2 (annealed at 800 ^oC in N_2). The full width at half maximum (FWHM) of the curve of Samp1 is 170" while that of Samp2 is 130". While the width is lower for the annealed sample, the wafer curvature is also greater for the as-bonded sample, partially offsetting the different widths. However, the broader shoulders observed below the half maximum intensity of the sample annealed at 800 ^oC shows that other, extended defects may evolve during the annealing. This behavior has been observed in implanted and annealed GaN.[32]

Figure 6(c) shows the comparison of calculated and measured thermal conductivity. The four samples studied in this work were heated to 450 ^oC during the exfoliation transfer process. As noted above, Samp2 and Samp4 were further annealed at 800 ^oC in N_2. The 450 ^oC heating primarily drives the H ions out of β-Ga_2O_3 films while the high temperature annealing (800 ^oC) reduces the amounts of implantation-induced point defects in the films. Phonons scatter with defects due to mass difference, and the local strain which induces changes in local bond stiffness, lattice constant, and Gruneisen parameter.[33] If we do not consider mass difference in the H ion

implanted β-Ga_2O_3 after H ions are driven out by the 450 ºC heating, then the local strain field is the primary source for phonon-defect scatterings, similar to other implanted materials.[33] The strain-induced phonon-defect scatterings reduce the thermal conductivity of these β-Ga_2O_3 thin films. The line labeled with "No strain" in Figure 6(c) shows the thickness dependent thermal conductivity which only considers the phonon-phonon scatterings and phonon-boundary scatterings. The measured thermal conductivity is slightly lower than the calculated thermal conductivity because of the remaining small amount of structural imperfection, as indicated by the shoulders in the XRD triple-axis rocking curve. The line labeled with "Strain" in Figure 6(c) shows the thickness dependent thermal conductivity which adds additional phonon-defect scattering associated with the implant-induced point defects. By adding the phonon-defect scattering into the calculation, we can match the measured thermal conductivity of the as-bonded samples to obtain the scattering cross-section (Γ_i) as 0.3. The phonon scattering mechanisms are shown in Figure 6(d). The strain field induced phonon-defect scattering exists in the films while the phonon-boundary scatterings occur at the film boundaries. Both the phonon-boundary scatterings and phonon-defect scatterings reduce the phonon mean free paths and the corresponding thermal conductivity of these thin films. The phonon-phonon scatterings are not included in the figure because they exist in both the films with strain and films without strains.

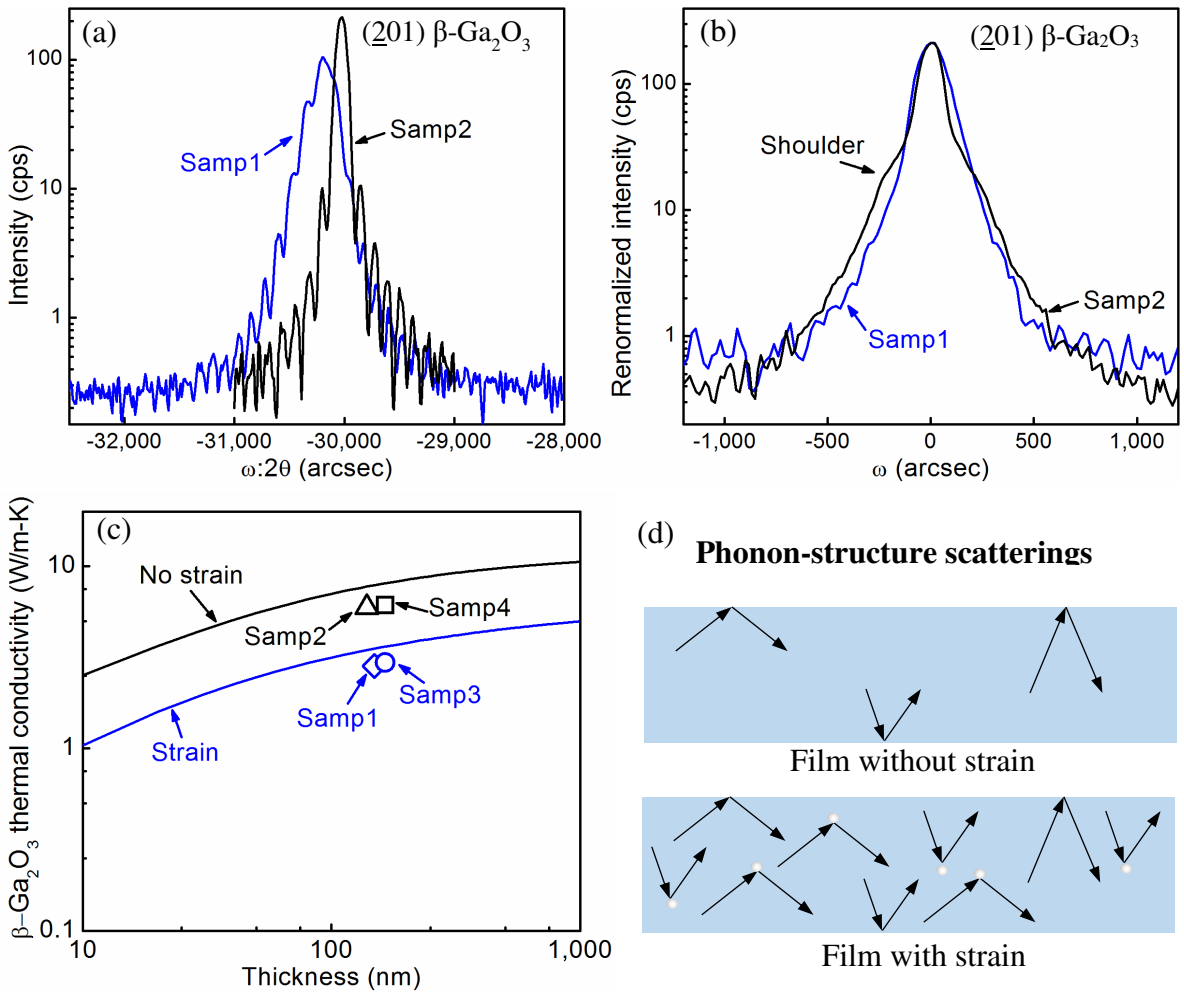


Figure 6. (a) Triple-axis (ω:2θ) scans of Samp1 and Samp2. (b) Triple-axis rocking curves of Samp1 and Samp2. The FWHM of Samp1 is 170" while that of Samp2 is 130". (c) Comparison of calculated and measured thermal conductivity of the monocrystalline β-Ga_2O_3 thin films. (d) Phonon scatterings with film boundaries and defects. The strain in the as-bonded samples induces phonon-defect scattering and correspondingly reduce thermal conductivity.

CONCLUSIONS

This work reports a scalable thermal management strategy to heterogeneously integrate wafer-scale monocrystalline β-Ga_2O_3 thin films on high thermal conductivity 4H-SiC substrates by the

ion-cutting and the surface activated bonding techniques. We observe reasonably high TBC values (67-100 MW/m^2-K) of the β-Ga_2O_3-SiC interfaces. The TBC values increase about 39% and 32% when decreasing the Al_2O_3 interlayer thickness from 30 nm to 10 nm for the as-bonded and annealed interfaces, respectively. The β-Ga_2O_3-SiC TBC is reduced slightly after the thermal annealing at 800 ^oC in N_2 which is possibly due to Ga element or GaO_x diffusion into the Al_2O_3 interlayer. A Landauer formula with DMM is used to calculate the β-Ga_2O_3-SiC TBC, the β-Ga_2O_3-Al_2O_3 TBC, and the Al_2O_3-SiC TBC. The calculated β-Ga_2O_3-SiC TBC are close to the measured ones. The β-Ga_2O_3 thermal conductivity increases by a factor of two after annealing in N_2 at 800 ^oC due to the removal of implantation-induced strain in the films. A Callaway model is used to understand the measured β-Ga_2O_3 thermal conductivity by considering the phonon scatterings with phonons, film boundaries, and strain. Additionally, fourteen different spots were measured to evaluate the uniformity. The small spot-spot variations of both TBC and β-Ga_2O_3 thermal conductivity confirm the uniformity and high-quality of the bonding, exfoliation, and annealing. Our work paves the way for thermal management of power electronics and β-Ga_2O_3 related semiconductor devices.

EXPERIMENTAL SECTION

Sample Preparation. The on-axis (0001) 4-inch 4H-SiC wafers with a thickness of ~500 μm were purchased from SICC Co., Ltd. The ($\bar{2}$01) 2-inch β-Ga_2O_3 wafers with a thickness ~680 μm were purchased from Novel Crystal Technology Inc. The Si-face of the 4H-SiC was used as the bonding surface. All of the bonding surfaces were polished by chemical mechanical polishing (CMP). The root-mean-square (RMS) surface roughness of the SiC and β-Ga_2O_3 surfaces are ~0.30 nm and ~0.27 nm, respectively.

The thin film exfoliation can be achieved by either hydrogen or helium ion implantation.[29] In the present study, the β-Ga_2O_3 wafers were implanted by hydrogen ions (H^+) with an energy of 35 keV and implantation dose around 1×10^{17} cm^{-2} at room temperature. The ion implantation was carried out with a 7-degree tilt to minimize the ion channeling effect.[34] The implanted H^+ ions accumulate in the β-Ga_2O_3 at 200-400 nm beneath the surface. After the ion implantation, an Al_2O_3 film was deposited on the β-Ga_2O_3 wafers by plasma-enhanced atomic layer deposition (PEALD) to avoid the surface damage layer during the following surface activation process for bonding and possibly blocking electrical breakdown due to the defective interfaces. Trimethylaluminum (TMA) and O_2 plasma were used as ALD precursors. Ar gas flow served as both carriers and purging gas. The O_2 plasma power was 100 W. The total flow rate of the Ar was 200 sccm (standard cubic centimeters per minute). The Al_2O_3 thin films were grown at 200 ^oC. The growth rate per cycle (GPC) was typically 1.3 Å/cycle for Al_2O_3. Al_2O_3 films with two kinds of thicknesses (30 nm for Samp1 and Samp2, and 10 nm for Samp3 and Samp4) were deposited on the implanted β-Ga_2O_3 wafers. After that, the two wafers were bonded together at room temperature by the surface-activated bonding (SAB) technique. The SAB machine consists of a load-lock chamber and a processing-bonding chamber. A silicon source is placed in the ion source to generate a mixture of Si ions and Ar ions. The small amount of Si ions added into the Ar ion beam was aimed to reduce the carbonation of SiC surface during the ion beam bombardment. The Ar ions activate the SiC and Al_2O_3 (on the β-Ga_2O_3) surfaces to create dangling bonds. After the surface activation, the samples were bonded directly at room temperature. The background pressure of the processing-bonding chamber is kept at 5×10^{-6} Pa. The ion beam source has a power of 1.0 kV and 100 mA.

During surface activation, the pressure of the processing-bonding chamber becomes ~4.0×10^{-1} Pa. Other bonding parameters can be found in the literature.[35]

By heating the bonded wafers at 450 °C, the implanted H^+ ions accumulate near the projected range of the ion implantation to induce the exfoliation of a β-Ga_2O_3 thin film.[28, 30-31, 34] After that, a β-Ga_2O_3 thin film with a thickness of less than 400 nm was split from the bulk β-Ga_2O_3 wafer and transferred to SiC substrate. The exfoliated β-Ga_2O_3 thin film was thinned by CMP to obtain a good TDTR sensitivity of the buried β-Ga_2O_3-SiC interface (100-200 nm thick). The post-exfoliated β-Ga_2O_3 bulk wafer can be repeatedly used for multiple exfoliations after polishing. The transferred 2-inch β-Ga_2O_3-on-SiC wafers were diced into 10 mm × 8 mm chips. Some chips were annealed in flowing N_2 gas. The temperature was ramped to 800 °C with a rate of 5 °C/min and then held at 800 °C for 30 min. Finally, a ~100-nm-thick Al layer was deposited on the surface of all samples as TDTR transducer.

TDTR Measurements. TDTR is a femtosecond-laser-based optical pump-probe technique to measure thermal properties of both bulk and nanostructured materials.[9, 20] A modulated pump beam heats the sample surface periodically while a delayed probe beam detects the surface temperature variation via thermoreflectance.[22] The variation of the surface temperature is fitted with an analytical heat transfer solution of the samples to infer the unknown thermal parameters.[10, 22] The measurements in this work are similar to those in references.[10-11, 36-37]

A 10 X objective (pump radius 10.1 μm and probe radius 5.8 μm) is used for the TDTR measurements with a modulation frequency of 2.2 MHz. The low modulation frequency is to

obtain deep thermal penetration depth to get enough sensitivity for the buried β-Ga_2O_3-SiC interfaces. More information about TDTR sensitivity can be found in the Supporting Information. A bare SiC wafer was measured first to obtain the SiC thermal conductivity (337 W/m-K), which was used as a known parameter in the data fitting of the measurements on β-Ga_2O_3 on SiC samples. The Al and β-Ga_2O_3 thicknesses were measured by the picosecond acoustic technique and confirmed with TEM and XRD. The thermal conductivity of Al is obtained by measuring the electrical conductivity and applying Wiedemann-Franz law. The heat capacity of the Al transducer, the β-Ga_2O_3 thin film, and the SiC substrates are from literature.[22, 26, 36, 38]

Materials Characterization. Cross-section TEM samples were prepared with a FEI Helios dual beam focused ion beam (FIB) system. The interfacial structures were characterized by a HR-STEM (Probe-corrected FEI Titan) and the interfacial composition was measured by EELS (Gatan Enfinium). The observation in this study is along the [11$\underline{2}$0] axis of SiC and the [010] axis of β-Ga_2O_3. Triple-axis x-ray diffraction (XRD) measurements were performed on a Bruker-JV D1 high-resolution X-ray diffractometer with incident beam conditioning that includes a parallel beam optical element and a (110) channel-cut silicon crystal that produces a monochromatic, highly collimated Cu $K\alpha_1$ beam. The scattered beam optics also included a (110) channel-cut silicon crystal. Both longitudinal (Δ (ω:2θ)) scans and rocking curves ($\Delta\omega$) were recorded under these triple axis HRXRD conditions.

Theoretical Modelings. A Landauer approach with DMM is used to calcuate the β-Ga_2O_3-Al_2O_3, Al_2O_3-SiC, and β-Ga_2O_3-SiC TBC. Because the transmission function from DMM does not depend on the angle of incidence, the Landauer formula is:

$$G = \sum_p \frac{1}{4} \int D_1(\omega) \frac{df_{BE}}{dT} \hbar\omega v_1(\omega) \tau_{12}(\omega) d\omega \,. \quad (1)$$

where D is the phonon density of states, f_{BE} is the Bose-Einstein distribution function, $\hbar$ is the reduced Planck constant, ω is the phonon angular frequency, v is the phonon group velocity of material 1, τ_{12} is the transmission coefficient from material 1 to 2, and the sum is over all incident phonon modes. The expression of the transmission function from DMM[25] is

$$\tau_{12}(\omega) = \frac{\sum_p M_2(\omega)}{\sum_p M_1(\omega) + \sum_p M_2(\omega)}, \quad (2)$$

where M is the phonon number of modes. The phonon properties of diamond are obtained from first-principle calculations with VASP.

A Callaway model is used to understand the measured β-Ga_2O_3 thermal conductivity by taking account of the film boundary scattering and strain effect in the films.[39-41] The phonon dispersion relation of β-Ga_2O_3 is calculated by density-functional-theory (DFT). All the phonon polarizations are considered (acoustic and optical, totally 30 branches for β-Ga_2O_3). The Callaway model gives an expression of thermal conductivity k as an integration in frequency space:

$$k = \frac{1}{3} \sum_p \int_0^{\omega_{\text{cut-off}}} \hbar\omega D_\lambda \frac{df_{BE}}{dT} v_\lambda^2 \tau_{C,\lambda}, \quad (3)$$

where the 1/3 comes from the isotropic assumption, $\sum_p$ is over all phonon polarizations, v_λ is the modal phonon group velocity, $\tau_{C,\lambda}$ is the modal combined relaxation time of phonon mode λ, D_λ is the modal phonon density of states, and f_{BE} is the Bose-Einstein distribution function. Three phonon scattering mechanisms are included in our calculation: phonon-phonon scatterings, phonon-boundary scatterings, and phonon-defect scatterings. The effective relaxation time $\tau_{C,\lambda}$ of each phonon mode can be obtained from the Matthiessen's rule[41]:

$$\tau_{C,\lambda} = \left(\frac{1}{\tau_U} + \frac{1}{\tau_D} + \frac{1}{\tau_B} \right)^{-1} \quad (4)$$

where τ_U, τ_D, and τ_B are the relaxation time of Umklapp phonon-phonon scatterings, phonon-defect scatterings, and phonon-boundary scatterings, respectively. The scattering rates can be shown below.[33, 42]

$$\frac{1}{\tau_U} = BT\omega^2 e^{-\frac{C}{T}}, \quad (5)$$

$$\frac{1}{\tau_D} = \frac{V\omega^4}{4\pi v^3}\Gamma_i, \quad (6)$$

$$\frac{1}{\tau_B} = v/d, \quad (7)$$

where B and C are constants (fitting parameters), V is the atomic volume of the β-Ga_2O_3, Γ_i is the parameter to describe the strength of phonon-defect scattering of defect i, and d is the thickness of the β-Ga_2O_3 film. When calculating the scattering rates of different mechanisms, the B (2.3×10^{-18} s/K) and C (130 K) are obtained by fitting with the previous experimental measurements on bulk β-Ga_2O_3 in which phonon-phonon scatterings dominate the thermal transport, especially at high temperatures.[8, 26]

Supporting Information: The supporting information includes additional STEM images of the bonded interfaces (Figures S1-S2), TDTR sensitivity (Figure S3), TDTR data fitting (Figure S4), picosecond acoustic echoes (Figure S5), and additional EELS maps (Figure S6).

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Author contributions: Z. C., J. S., and S. G. did the thermal measurements and calculations. F. M., T. Y., W. X., T. S., and X. O. fabricated the samples and finished part of the material characterizations. M. E. L., Y. W., K. H., and M. S. G. finished the XRD measurements and part of the TEM study. Z. C. wrote the manuscript with inputs from all other authors.

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